

MITHUN PARAMATHASAN

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Education

McMaster University

Sep 2021 – Apr 2026

B.Eng in Software Engineering

Hamilton, ON

- Golden Key Honours Scholar — **GPA: 3.55**
- **Relevant Courses:** Software Design, Data Structures and Algorithms, Advanced Databases, Applications of Machine Learning, Software Testing, Object-Oriented Programming

Work Experience

Government of Ontario – MPBSDP Ministry

Sep 2024 – Aug 2025

Software Developer Co-op

Toronto, ON

- Analyzed **13 legacy Ontario Student Assistance Program (OSAP) applications** and documented functionality across **30+ business modules**, contributing to **Request for Proposal (RFP)** documentation for a **\$20M modernization initiative** involving technical dependencies and migration planning
- Advanced remediation planning for **7M+ lines** of legacy **RPG, COBOL, and LANSA/SYNON** code, supporting migration from **IBM AS400** mainframe infrastructure to scalable **Java microservices** architecture
- Collaborated with Google Cloud AI, Microsoft Azure, and IBM teams to assess **LLM-driven** solutions for legacy code analysis, automated documentation generation, and transpiler feasibility

CERC@MARC

May 2023 – Sep 2023

VR Developer Co-op

Hamilton, ON

- Optimized a **Unity-based** train simulation by developing **C#** animation scripts and component interaction systems, and integrating pre-built assets through **Virtualware's Viroo VR platform**, to improve VR scene frame rates by **20%**
- Implemented and unit tested immersive train animations for **Oculus Rift** and **HTC Vive** headsets to ensure smooth performance during client demo reviews

Projects

Capstone Project – Pocket AI | *Demo Video*

React Native | TypeScript | Python | Flask | PostgreSQL

- Engineered a voice-driven task management mobile application for an industry-sponsored project with **CATTLEytics**, leveraging **Whisper AI** for speech transcription and **Groq LLM** for context-aware task extraction in farm settings
- Integrated with the CATTLEytics platform for cloud-based task management, and used local **PostgreSQL** persistence and asynchronous synchronization workflows to improve reliability during third-party API failures

GymBuddy | *Source Code*

Next.js | TypeScript | Node.js | MongoDB

- Architected a full-stack fitness tracking application with a **Next.js** frontend and **Node.js** backend, implementing workout planning, and progress visualization features with **MongoDB**-based user data storage
- Integrated and filtered **USDA FoodData API** results to improve nutritional search relevance

Chest X-Ray Pneumonia Detection | *Source Code*

PyTorch | scikit-learn | Python

- Built a chest X-ray pneumonia detection system using **PyTorch** and **scikit-learn**, implementing **ResNet18** feature extraction and CNN classification, achieving **88%** test accuracy and a **0.91 F1-score** on 5,800+ medical images

Clubs and Organizations

Open-Source Core Team Member | *Source Code*

Sep 2023 – Apr 2024

- Developed a full-stack gamified learning platform for the **McMaster Google Developer Student Club**, implementing interactive quizzes, lessons, leaderboards and user progress tracking using **Flask, JavaScript, and SQLAlchemy**
- Implemented secure authentication, **RESTful APIs**, and optimized database operations, reducing average page load times by **25%** during local testing
- Containerized the application with **Docker** and configured cloud deployment workflows on **Google Cloud Platform (GCP)** with **Cloud SQL** integration

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML5, CSS, Golang, C

Developer Tools: Git, Postman, PostgreSQL, MongoDB, Docker, GitHub Actions, Google Cloud Platform

Libraries/Frameworks: React.js, Next.js, Node.js, Express.js, Flask, Tailwind CSS, Framer Motion, JUnit, Jest